

Things I Don't Understand About Quantum Error Correction

Tree-based vs surface codes: questions that only make sense inside the frame

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Poster Session: Wed 26 Aug, 16:30–18:00 · Rowan Brad Quni-Gudzinas · QNFO

Panel 1 — Why this poster exists

I am an independent researcher who came to QEC from ultrametric geometry and the energy cost of computation. This poster is a notebook of confusions treated as data.

Every question below shares one property: **it makes no sense outside the frame it questions**. That is deliberate: where a question becomes nonsense is exactly where a definition is doing the work.

The method, in four steps:

- State the construct in its own terms — surface codes, thresholds, logical qubits, decoders.
- Ask the question I am not supposed to need to ask, exactly as a naive outsider would.
- Watch where the question stops parsing — that locale is the seam.
- Treat the seam as data: the construct carried more than its proof.

Correct me, and the poster has done its job. I am not claiming the field is wrong; I am asking where the pieces were checked against each other. If the constructs were compatible, the naïve questions would answer themselves.

Panel 3 — Questions 1–4

1 • What is a logical qubit, physically?

Outside the frame: "logical" is not a physical predicate — the question is nonsense.

Inside, the answer is a definition: a logical qubit is defined as what the code protects. The code defines the thing it protects; the definition is the protection.

2 • Why does a surface code need a surface?

Outside the frame: a code is a set of states; no geometry is required.

Inside, the answer is a definition: the logical operators are defined by the boundary. The surface defines the code that defines the surface — the geometry is the definition.

3 • Where is the error in the error model?

Outside the frame: an error is an error; the question is empty.

Inside, the answer is a definition: the noise model defines which errors exist. Correlated errors are defined out of the frame, then declared absent. The definition is the measurement.

4 • What does the threshold guarantee?

Outside the frame: a number guarantees nothing.

Inside, the answer is a definition: the threshold is a crossing point inside a model family. The guarantee holds inside the model that defines it, and is undefined outside it.

Panel 5 — What would change my mind

A matched-noise comparison: the tree family vs a surface code under the same correlated-noise model, in classical simulation. If the advantage falls to parity, the comparison on this poster is answered.

A decoder reality check: does a known decoder family already cover tree-structured codes, and how does its complexity scale?

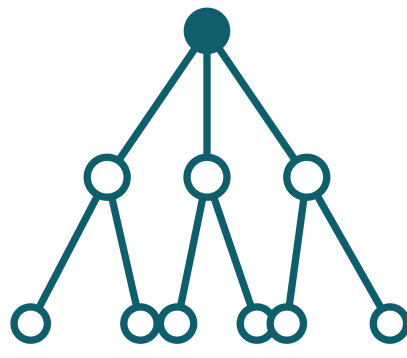
An independent re-run: the simulator and the protocol are handed over to anyone who wants to run the comparison themselves.

Perfect tensors for $p > 2$: the binary $[[3,1,1]]$ case is settled; larger primes are open.

What is not claimed: no hardware result, no fault-tolerance claim under circuit-level noise, no advantage under independent errors — the qudit family sits at $\approx 2 \times 10^{-4}$ there, below surface codes. The claim is narrow: under correlated failure, the tree geometry may need fewer active interventions, and the matched comparison decides it.

Numeric criteria: matched-noise parity $\Rightarrow$ the advantage claim is withdrawn · a decoder that already covers trees $\Rightarrow$ ask 2 answered · an independent re-run that fails $\Rightarrow$ the simulation claim is withdrawn. Every criterion is stated with its number.

Panel 2 — The comparison: tree-based vs surface codes



One instance of the class: the Bruhat–Tits tree T_2 .
The class: nested, distinction-based ultrametric trees.

Surface codes (the standard): qubits on a 2D lattice; active syndrome measurement; a classical decoder decides what to correct. The code is defined by its boundary.

Tree-based codes (the alternative): nested, distinction-based hierarchical structures — ultrametric trees; the Bruhat–Tits tree T_2 is one instance, not the concept. A $[[3,1,1]]$ perfect tensor at every internal vertex; protection from concatenation across the tree, not from active loops.

All numbers are classical simulation. No tree code has run on hardware.

Error model (qubit, code-capacity)	Tree	Surface	Ratio
Bit-flip	50.0%	~10.9%	4.6×
Depolarizing	75.0%	~1.0%	75×
Independent X+Z	17.30%	~18.9%	0.92×

Depolarizing uses the code's structured-noise channel (not yet published); surface comparators are standard published figures. Under independent errors the qudit generalization sits at $\approx 2 \times 10^{-4}$ — about 55× below surface codes; the qubit claim targets correlated-failure regimes.

Panel 4 — Questions 5–8

5 • Why must correction be a loop?

Outside the frame: correction needs no loop.

Inside, the answer is a definition: QEC is defined as measure $\rightarrow$ classical decoder $\rightarrow$ feedback. A code that corrects by geometry alone is outside the definition — which is the claim under test.

6 • What is one logical qubit worth?

Outside the frame: worth is not a code property.

Inside, the answer is a definition: the rate $n:k$ is a definition of the family. The cost stays a definition until it is priced in joules per correct answer.

7 • What does distance $d = 1$ mean?

Outside the frame: a "code" of distance 1 corrects nothing — it is not a code.

Inside, the answer is a definition: d is the minimum logical-operator weight; $d = 1$ means "corrects nothing by definition" — and concatenation across the tree turns the definition into protection. The tree is where the definition stops being the answer.

8 • Is the decoder part of the code?

Outside the frame: malformed — a decoder is a decoder.

Inside, the answer is a definition: the quantum/classical boundary is drawn by convention, and the decoder's energy is priced nowhere. The convention is the frame; the frame is the cost.

Panel 6 — The statement, the caption, the seams

The one-line statement: "My confusions are the fault lines: where a question makes no sense outside the frame, the frame is answering with a definition."

The caption: the naive question is the probe. I am not claiming the field is wrong — I am asking where the pieces were checked against each other. *If the constructs were compatible, the naive questions would answer themselves.*

Seam 1 — Discrete abstraction: QEC defends the discreteness of the circuit model; the choice of discreteness is never priced.

Seam 2 — Logical qubit: defined as what the code protects; the definition is the protection.

Seam 3 — Noise model: the model defines which errors exist; thresholds live inside the model.

Seam 4 — Classical control: the decoder is classical; the loop's energy is priced nowhere.

Seam 5 — Cost currency: T-count is countable, so it is counted; joules are measurable, and not measured.

Seam 6 — Geometry: Euclidean 2D assumed; the tree geometry unexamined. The staircase signature I could not find examined anywhere.